



ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 4: HealthConnect Project Kickoff & Problem Understanding

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 4
Project	HealthConnect Clinic Experience Lab
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 4 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 4 Submission Form

1. BUSINESS SCENARIO

HealthConnect Clinic is a fictional healthcare provider that manages appointment-based services.

The clinic is experiencing several operational and patient support challenges, including:

- Patients missing scheduled appointments.
- Difficulty understanding the factors associated with appointment no-shows.
- Inefficient use of appointment slots when patients fail to attend.
- Repetitive patient enquiries relating to appointments and clinic procedures.
- The need to improve patient engagement and administrative support.

HealthConnect Clinic wants to explore how data, machine learning, and Generative AI can support better decision-making and improve the patient experience.

Central Project Question

How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

From Week 4 onwards, all interns will contribute to this shared project from the perspective of their respective professional tracks.

2. PROJECT TRANSITION AND CONTINUITY

During Weeks 1–3, you completed tasks designed to develop and strengthen your practical skills within your respective internship tracks.

The HealthConnect Experience Lab begins in Week 4.

Please note:

- You are not required to redo or replace your Week 1–3 tasks.
- Your previous work remains part of your internship learning journey.
- From Week 4 onwards, you will work on the HealthConnect project.
- Each subsequent week will build on the progress made in the previous week.
- The goal is to develop a structured and portfolio-ready project by the end of the Experience Lab.

Your project journey will progress through the following stages:

Problem Understanding
Analysis and Solution Design
Development
Testing and Refinement
Final Presentation

3. PROJECT RESOURCES

The following resources have been provided for the HealthConnect Experience Lab.

3.1 HealthConnect Appointment Dataset

File: HealthConnect_Appointment_Data.csv

Download Link: [Click here to download the HealthConnect Appointment Dataset](#)

This dataset contains fictional and anonymized appointment records.

It includes information relating to:

- Patient demographics.
- Appointment details.
- Booking information.
- Previous appointment history.

- Previous no-shows.
- Reminder information.
- Distance to the clinic.
- Waiting time.
- Appointment outcomes.

Primary Users:

- Data Analytics
- Data Science
- Machine Learning Engineering

3.2 HealthConnect Data Dictionary

File: HealthConnect_Data_Dictionary.xlsx

Download Link: [Click here to download the HealthConnect Data Dictionary](#)

The Data Dictionary provides explanations of the variables contained in the appointment dataset.

Primary Users:

- Data Analytics
- Data Science
- Machine Learning Engineering

3.3 HealthConnect Clinic Knowledge Base

File: HealthConnect_Clinic_Knowledge_Base.docx

Download Link: [Click here to download the HealthConnect Clinic Knowledge Base.docx](#)

This document contains approved fictional information about HealthConnect Clinic, including:

- Clinic information.
- Locations and opening hours.
- Available services.
- Appointment procedures.
- Rescheduling and cancellation procedures.
- Frequently asked questions.
- AI assistant safety rules.
- Escalation guidelines.

Primary Users:

- Generative AI

Important Resource Instructions

- Do not overwrite or alter the original project resources.
- Save any cleaned, transformed, or processed data separately.
- Document significant changes made during your work.
- Use the resources relevant to your assigned track.
- Generative AI solutions must remain within the approved scope of the Knowledge Base.
- Do not invent clinic or medical information.

4. ASSIGNMENT OBJECTIVE

The objective of Week 4 is to establish a clear understanding of the HealthConnect business problem and define how your professional track will contribute to the overall project.

By the end of the week, you should be able to:

- Explain the HealthConnect business problem.
- Understand the resources relevant to your track.
- Define your track-specific problem or objective.
- Develop an initial approach to your work.
- Identify key assumptions, limitations, risks, or dependencies.
- Document the foundation for the next phase of the project.

5. TOOLS AND TECHNOLOGIES

Use tools appropriate to your internship track and proposed approach.

You are not required to use every tool listed below.

Project Management

Tools include:

- Microsoft Excel or Google Sheets
- Microsoft Word or Google Docs
- PowerPoint or Google Slides
- Trello
- Notion
- Jira or similar project management tools

Data Analytics

Tools include:

- SQL
- Python
- Power BI
- Tableau

Data Science

Tools include:

- Python
- Jupyter Notebook or Google Colab
- VS Code
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Machine Learning Engineering

Tools include:

- Python
- Jupyter Notebook or VS Code
- Git
- GitHub
- Appropriate machine learning libraries

Generative AI

Tools include:

- Python
- Jupyter Notebook or VS Code
- Git and GitHub
- Appropriate Large Language Model tools, APIs, or frameworks where applicable

Note: Week 4 focuses on understanding, planning, and initial exploration. You are not expected to complete the final solution, deploy a model, build a complete dashboard, or develop a fully functional AI assistant at this stage.

6. ASSIGNMENT TASKS

PART 1: PROJECT AND RESOURCE REVIEW

Applicable to All Tracks

Review:

- The HealthConnect business scenario.
- The project objective.
- The resources relevant to your internship track.

Your review should help you clearly define:

1. The problem you will address.
2. Your role within the broader HealthConnect project.
3. The resources you will use.
4. Your initial approach to the project.

You do not need to prepare a separate report for this part. Your findings should be incorporated into your track-specific Week 4 work.

PART 2: TRACK-SPECIFIC ASSIGNMENT

A. PROJECT MANAGEMENT TRACK

Your responsibility is to establish the initial structure for managing the HealthConnect project.

Tasks

1. Define the project scope and objectives.
2. Identify key stakeholders and project workstreams.
3. Develop a Work Breakdown Structure.
4. Create a high-level project timeline.
5. Identify major project risks and dependencies.
6. Develop an initial communication approach.

Week 4 Output

Prepare a Project Initiation Package containing:

- Project Brief or Charter.
- Scope Statement.
- Stakeholder Register.
- Work Breakdown Structure.
- High-Level Timeline.
- Initial Risk and Dependency Register.
- Communication Approach.

B. DATA ANALYTICS TRACK

Your responsibility is to understand the appointment data and identify how it can be used to investigate appointment attendance and no-show patterns.

Tasks

1. Review the Data Dictionary and dataset structure.
2. Conduct an initial data quality assessment.
3. Identify important variables relevant to appointment attendance and no-shows.
4. Define relevant business questions.
5. Propose meaningful KPIs.
6. Develop an initial analysis approach.

Week 4 Output

Prepare an Initial Analysis Document or Notebook containing:

- Dataset overview.
- Data quality assessment.
- Relevant business questions.
- Identify 3–5 potential KPIs that may be relevant to the business questions you intend to explore in subsequent stages of the project. Each potential KPI should be linked to a business question. At this stage, you are only expected to identify and justify the relevance of the KPIs; you are not required to calculate, analyse, or build visualisations for them.
- Initial analysis approach.

C. DATA SCIENCE TRACK

Your responsibility is to define the machine learning problem and assess whether the available data can support a no-show prediction solution.

Tasks

1. Review the dataset and relevant variables.
2. Assess the quality and suitability of the data.
3. Define the machine learning problem.
4. Identify the proposed target variable.
5. Identify potential input features.
6. Define how appointment outcomes such as cancellations will be handled.
7. Develop an initial modelling approach.

Week 4 Output

Prepare a Machine Learning Problem Definition Document or Notebook containing:

- Problem definition.

- Initial data assessment.
- Proposed target variable.
- Potential features.
- Initial modelling approach.
- Key modelling considerations.

D. MACHINE LEARNING ENGINEERING TRACK

Your responsibility is to design the initial structure of a reliable and reproducible machine learning solution.

Tasks

1. Define the expected input and output of the prediction system.
2. Identify the major stages of the machine learning workflow.
3. Develop a high-level system architecture.
4. Identify key data, model, and system dependencies.
5. Consider reproducibility, version control, and model management requirements.
6. Develop an initial implementation approach.

Week 4 Output

Prepare an ML System Design Document containing:

- ML problem overview.
- Input and output definition.
- High-level system architecture.
- Proposed ML workflow.
- Key dependencies.
- Initial implementation approach.

E. GENERATIVE AI TRACK

Your responsibility is to define how a Healthcare Information Assistant could provide useful administrative and informational support while operating safely.

Tasks

1. Identify the intended users of the assistant.
2. Define the assistant's primary use cases.
3. Define the questions or requests the assistant should support.
4. Define requests that fall outside its approved scope.
5. Establish safety boundaries and escalation scenarios.
6. Develop an initial assistant workflow or solution design.

Week 4 Output

Prepare an AI Assistant Design Document containing:

- Assistant purpose.
- Target users.
- Supported use cases.
- Out-of-scope requests.
- Safety boundaries.
- Escalation approach.
- Initial assistant workflow or solution design.

PART 3: ASSUMPTIONS, LIMITATIONS, RISKS AND DEPENDENCIES

Identify the factors that may affect your proposed approach.

Depending on your track, these may include:

- Data limitations.
- Missing or incomplete information.
- Technical constraints.
- Project risks.
- System dependencies.
- Modelling considerations.
- Ethical or safety concerns.

Include these within your track-specific Week 4 output.

You do not need to create a separate Assumptions, Limitations and Risk Summary.

PART 4: WEEK 4 PROJECT SUMMARY

Prepare a concise Week 4 Project Summary covering:

1. The specific problem your track will address.
2. The resources you used.
3. Your key observations from Week 4.
4. Your proposed approach.
5. The key considerations that may affect the project.
6. Your proposed focus for Week 5.

This summary should be concise and should complement—not repeat—your main project documentation.

7. SUBMISSION REQUIREMENTS

Your Week 4 submission should include:

1. Track-Specific Week 4 Output

Submit the required output for your internship track.

2. Week 4 Project Summary

Submit a concise summary of your Week 4 work and proposed Week 5 focus.

3. Supporting Files

Include any relevant files used to complete your work, such as:

- Notebooks.
- Reports.
- Project plans.
- Registers.
- Architecture diagrams.
- Workflow designs.
- Supporting documentation.

4. Google Drive Folder

Upload your relevant Week 4 files to your designated Google Drive folder.

Ensure that:

- The correct files are uploaded.
- Files are properly named.
- Your work is organised.
- Your assigned mentor has the required access.

5. GitHub Repository

Where applicable, update your existing GitHub repository with your Week 4 work.

Ensure that:

- Relevant files are included.
- Your project structure is organised.
- Your `README.md` is updated.
- Links and files are accessible.

8. PROFESSIONAL DEVELOPMENT REQUIREMENT

As part of your professional development, you should:

- Document your analytical, technical, or project process.
- Maintain an organised project folder or repository.
- Update your existing GitHub repository where applicable.
- Update your `README.md` to reflect your Week 4 project progress.
- Present your work professionally.
- Share meaningful learning or project progress on LinkedIn and X where appropriate.
- Tag **AnalystLab Africa** via LinkedIn.
- Tag **analystlabafric** via X.
- Use **#AnalystLabAfrica** where appropriate.

9. PROFESSIONAL BRANDING REMINDER

AnalystLab Africa encourages interns to document their learning journey professionally.

Sharing meaningful insights from your projects can help you:

- Build your portfolio.
- Demonstrate practical skills.
- Showcase your problem-solving approach.
- Document your professional growth.
- Increase your visibility to recruiters and employers.

Ensure that any work shared publicly accurately reflects your actual progress and learning.

Assessment Criteria

Assessment Criteria	Weight
Understanding of the Track-Specific Problem	20%
Review and Use of Relevant Project Resources	15%
Quality of Initial Analysis, Planning or Solution Design	25%
Identification of Assumptions, Limitations, Risks or Dependencies	15%
Documentation and Organisation	15%
Professional Presentation	10%
Total	100%

IMPORTANT REMINDER

Week 4 is the foundation stage of the HealthConnect Experience Lab.

You are not expected to complete the entire solution this week.

Your focus should be on:

- Understanding Your Problem
- Reviewing Relevant Resources
- Defining Your Approach
- Preparing for Project Development

Your Week 4 work should provide a clear and organized starting point for the activities that will continue in Week 5.